TIME AND SPACE COMPLEXITY

SUM OF FIRST N NATURAL NUMBERS

$S = 1 + 2 + 3 + 4 + 5 \ldots - \ldots + N$

$S = N + (N-1) + (N-2) \ldots \ldots + 2 + 1$

$2S = (N+1) + (N+1) + (N+1) \ldots \ldots + (N+1)$

$2S = N(N+1)$

$$S = \frac{N(N+1)}{2}$$

$\frac{N(N+1)}{2}$

$1 + 2 + 3 + 4 \ldots + 50$

$= \frac{50 \times 51}{2} = __$

$N = 10^9$

$$(1+2+3 \ldots +6) + 7 + 8 + 9 + 10 \ldots + 100$$

$$- (1+2+3+4+5+6)$$

$$= \frac{100 \times 101}{2} - \frac{6 \times 7}{2}$$

$$(1+2+3 \ldots + (L-1))$$

$$L + (L+1) + (L+2) \cdots + R$$

$$- (1 + 2 + 3 \cdots + (L-1))$$

$$= \frac{R(R+1)}{2} - \frac{(L-1)L}{2}$$

COUNT ELEMENTS IN A RANGE



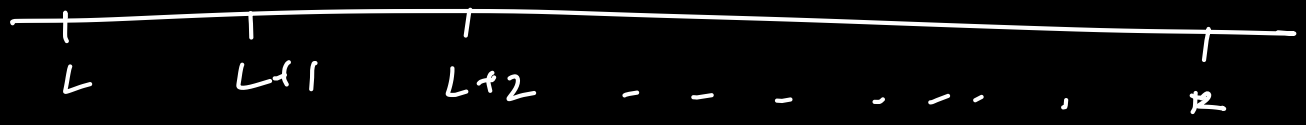
→ 75



→ $40 - 29 + 1$

$= 41 - 29$

$= 12$



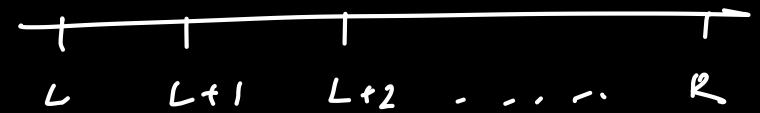
$R - L + 1$

$[L, R] \rightarrow R - L + 1$

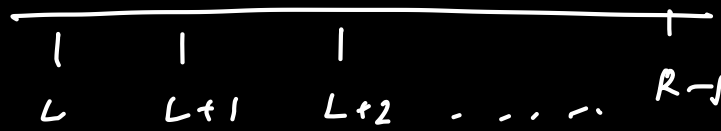
$(L, R] \rightarrow R - L$

$[L, R) \rightarrow R - L$

$(L, R) \rightarrow R - L - 1$







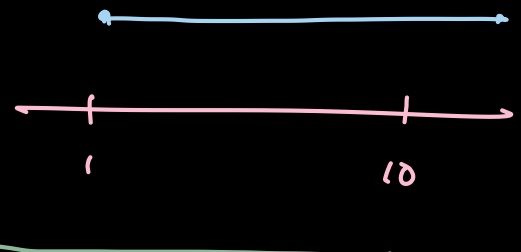
$L = R$

$[10, 10] \rightarrow 1$ (≥ 10 and ≤ 10)

$(10, 10] \rightarrow 0$ (> 10 and ≤ 10)

$[10, 10) \rightarrow 0$ (≥ 10 and < 10)

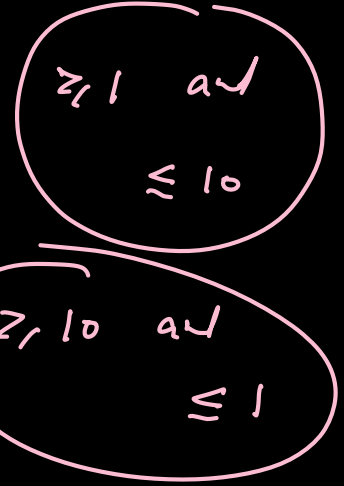
$(10, 10) \rightarrow 0$ (> 10 and < 10)



L > R

[1, 10] → 10

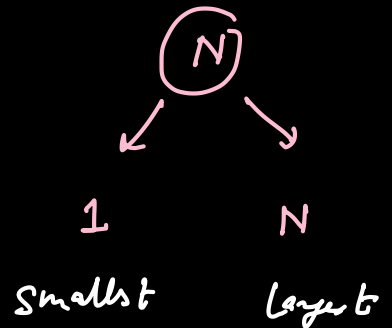
[10, 1] → 0



FACTORS

12 : 1, 2, 3, 4, 6, 12

20 : 1, 2, 4, 5, 10, 20




```
For( i = 1;  i <= N;  i++)
{       if( i is a factor of N)        N % i == 0
        {
            print (i)
        }
}
```

$1 \le N \le 10^{19}$

```
cnt = 0
for( i = 1; i <= N; i++)
{       if( i is a factor of N)
        {
            cnt ++
        }
}
```

→ N y. i == 0

loop runs N times

N iterations

↓

N iterations

CF / LC / CC ...

Standard

1 sec → 10^8 iterations

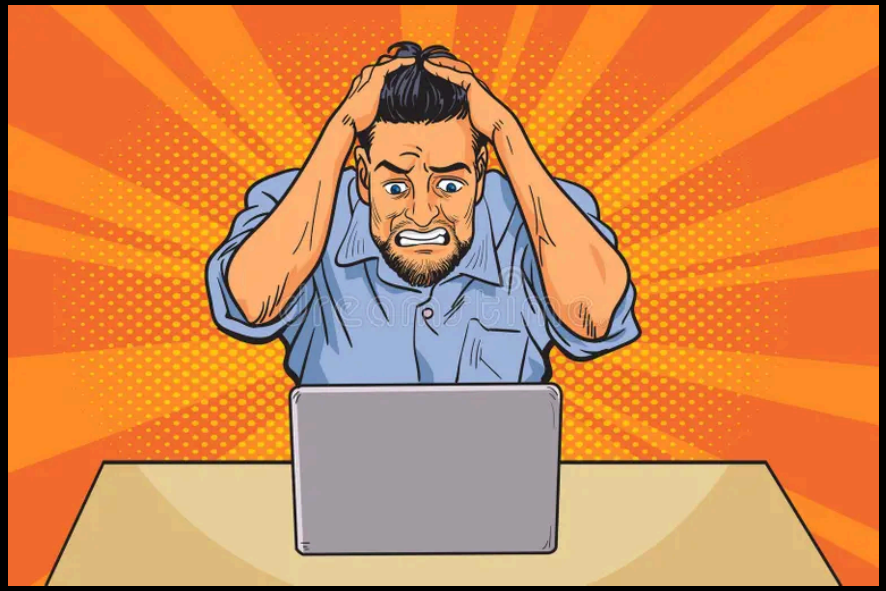
Time Limit Exceeded

N	itrs	time
10^8	10^8	$\frac{10^8}{10^8}$ = 1 sec
2×10^8	2×10^8	$\frac{2 \times 10^8}{10^8}$ = 2 secs
10^9	10^9	$\frac{10^9}{10^8}$ = 10 secs → 317 years
10^{18}	10^{18}	$\frac{10^{18}}{}$ = 10^{10} secs → mins, hours, days

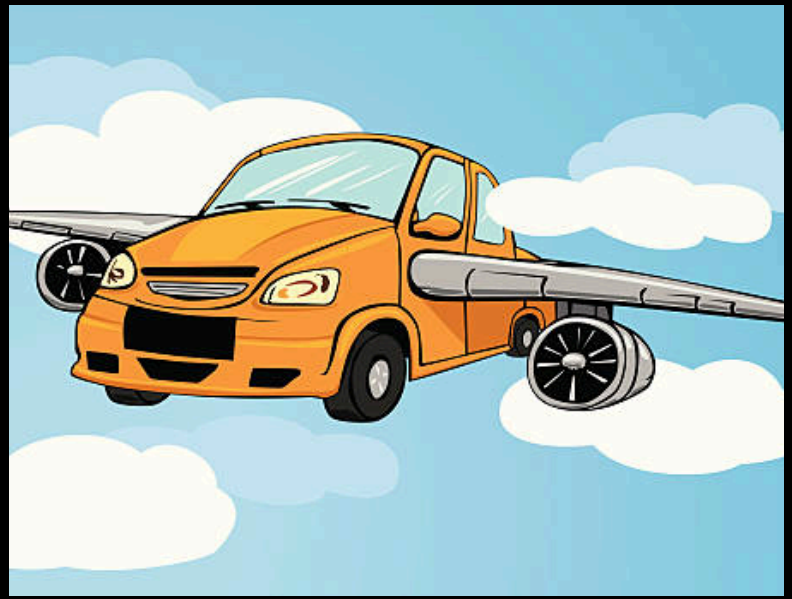
16^8

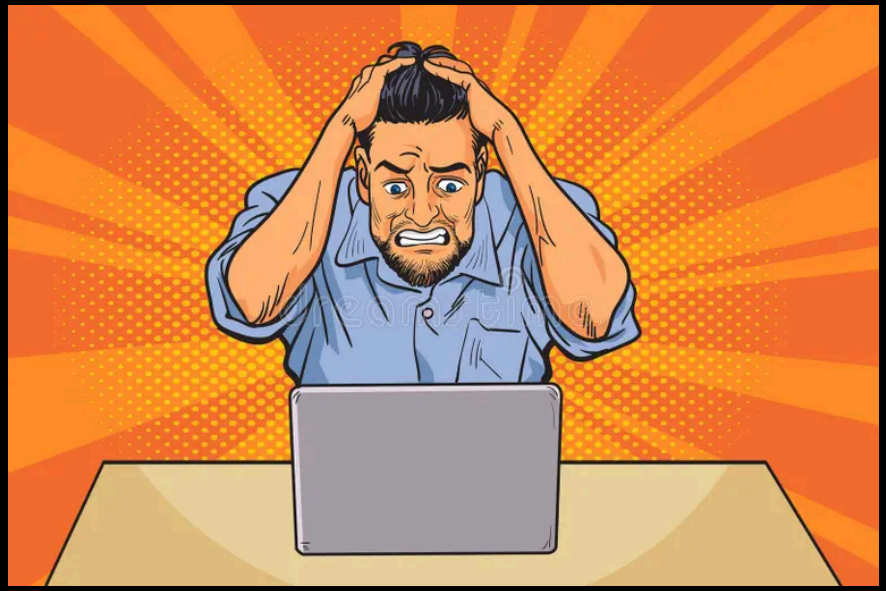
WHAT COULD HAPPEN IN 300 YEARS ?





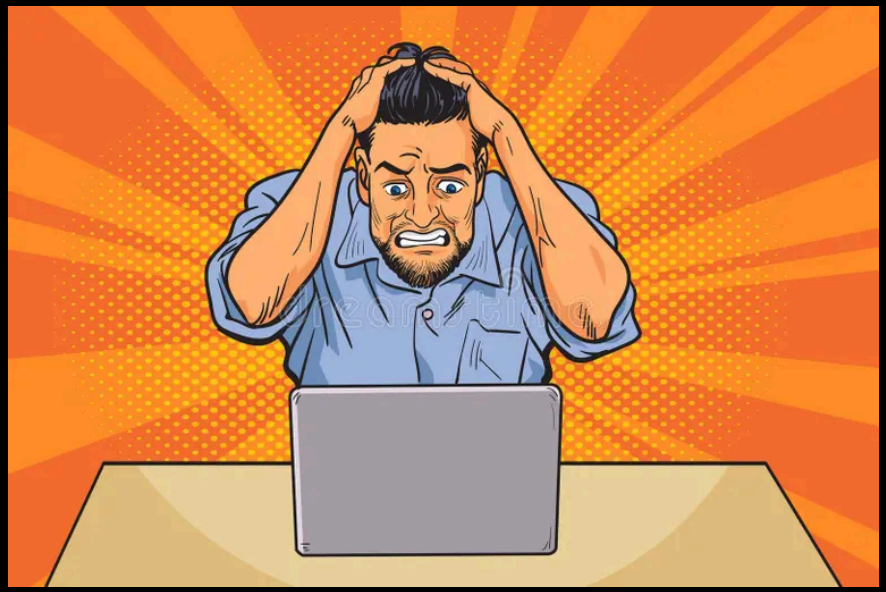
WHAT COULD HAPPEN IN 300 YEARS ?





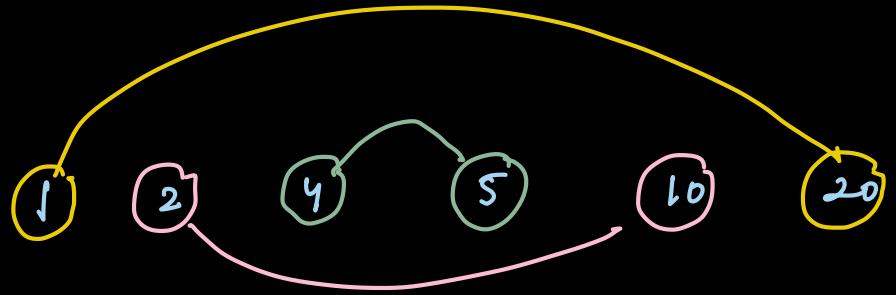
WHAT COULD HAPPEN IN 300 YEARS ?





OPTIMISING THE SOLUTION

N : 20



i		
1	1, 20	cnt += 2
2	2, 10	cnt += 2
3		
4	4, 5	cnt += 2
5		
6		
7		
8		

1 x 20 = 20

2 x 10 = 20

4 x 5 = 20

9

10

11

12

13

14

15

16

17

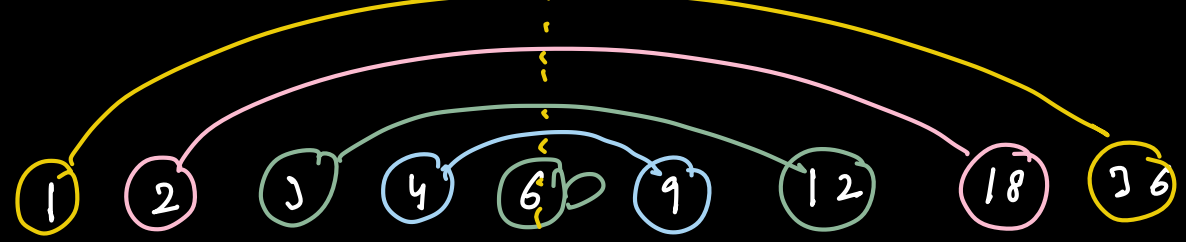
18

19

20



N : 36



i	f1	f2	
1	1,	36	cnt += 2
2	2,	18	cnt += 2
3	3,	12	cnt += 2
4	4,	9	cnt += 2
5			
6	6,	6	cnt += 1
7			
;	9,	4	
;			
;			

f1 ≤ f2

f1 = i

f2 = N / i

→ Small region

f1 ≤ f2

i ≤ N / i

i² ≤ N

stop (√N)

f1 ≥ f2

12, 1

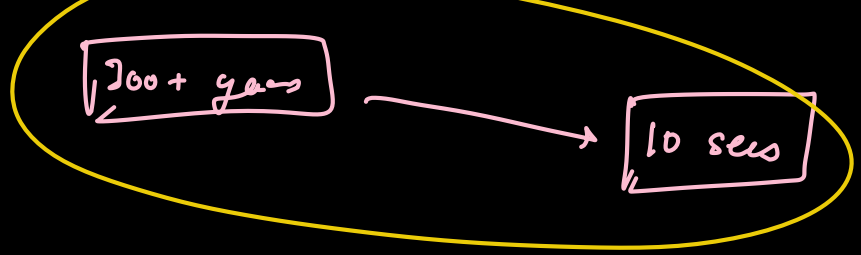
18, 2

36 36, 1

$i \leq \sqrt{N}$

```
cnt = 0
for( i = 1;  i <= Math.sqrt(N) ; i++)
{
    if ( N % i == 0)
    {
        f1 = i
        f2 = N / i
        if( f1 == f2)
        {
            cnt++
        }
        else
```

```
        {
            cnt += 2
        }
    }
}
print(cnt)
```



$(\sqrt{N})$

N	iters	time
10^8	10^4	$\frac{10^4}{10^8} = 10^{-4}$ secs < 1 sec
10^{16}	10^8	$\frac{10^8}{10^8} = 1$ sec
10^{18}	10^9	$\frac{10^9}{10^8} = 10$ secs

Prime → Exactly 2 factor → (1), (N)

(N)

1 → 1

2 → 2

3 → 2

4 → 3

5 → 2

6 → 4

7 → 2

8 → 4

9 → 3

256

128

$\sqrt{256}$

COUNT ITERATIONS

$N \geq 0$

$N < 10^{18}$

```
int i = N;

while(i > 1)
{
    i = i / 2;
}
```

$\log_2(N)$

$O(\log_2 N)$

$i = i / 6$

$\log_6(N)$

64

$N = 32$

$i = 32$ 2^5

16 2^4

8 2^3

4 2^2

2 2^1

1 2^0

$\sqrt{121} = 11.$

$N \times$

$\sqrt{N}$

$\sqrt{32} = 5.$

$\log_2(N)$

```
int i = N;

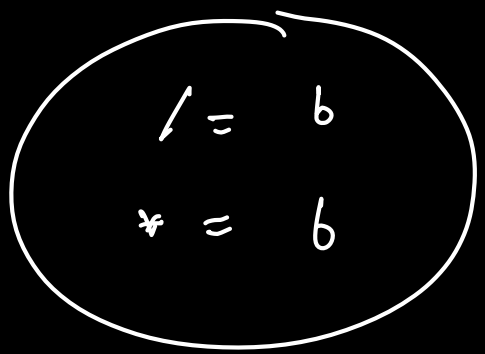
while(i > 1)
{
    i = i / 2;
}
```

COUNT ITERATIONS

```
int i = 1;

while(i <= N)
{
    i = i * 2;
}
```

$O(\log_2(N))$



N = 32

i = 1

↓

2

↓

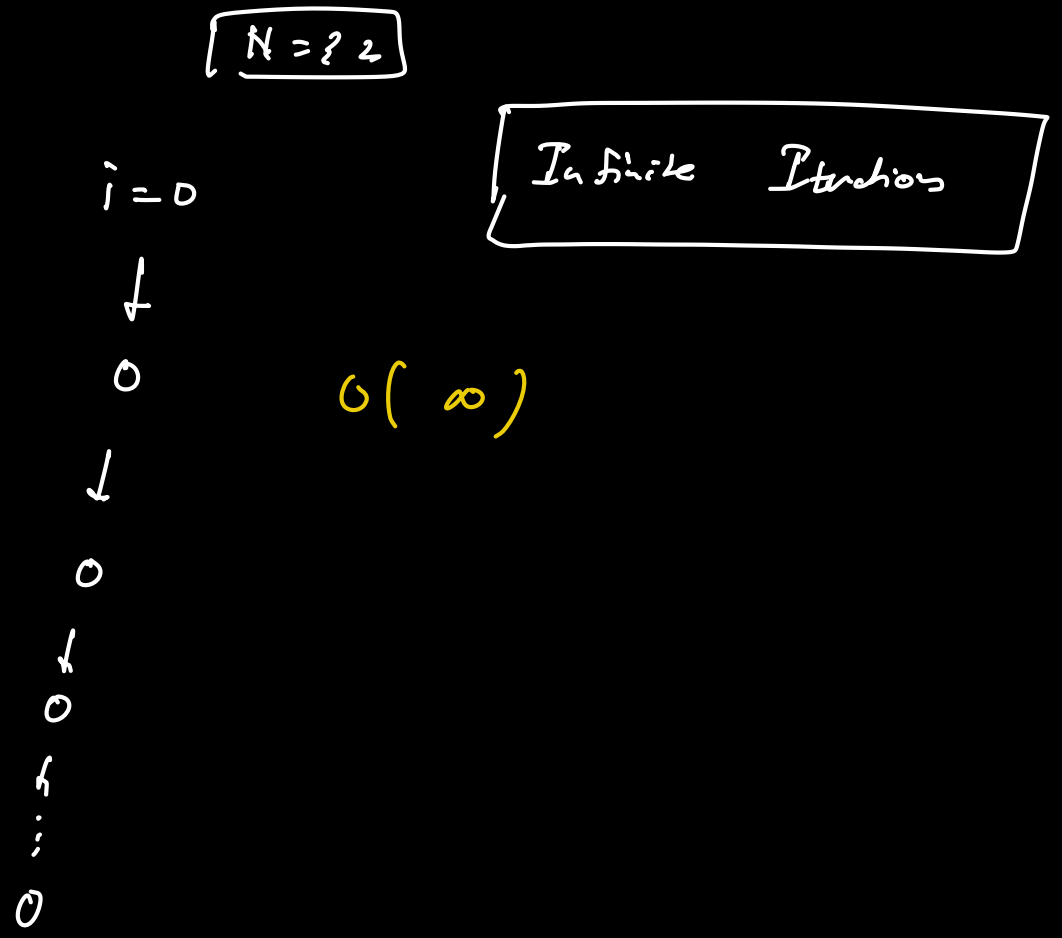
4 → 8 → 16 → 32 → 64

$\log_b N$

COUNT ITERATIONS

```
int i = 0;

while(i <= N)
{
    i = i * 2;
}
```



COUNT ITERATIONS

```
for(i = 1; i <= N; i++)
{
    for(j = 1; j <= N; j++)
    {
        print(i + j);
    }
}
```

$N(N+1)$

$O(N^2)$

i	j	# itr
1	1 to N	1 to N
2	"	"
3	"	"
⋮		
N	"	"

$N(N+1)$

COUNT ITERATIONS

```
for(i = 1; i <= N; i++)
{
    for(j = 1; j <= i; j++)
    {
        print(i + j);
    }
}
```

i	j	# itr
1	1 to 1	1
2	1 to 2	2
3	1 to 3	3
⋮	⋮	⋮
N	1 to N	N

$$N + \frac{N(N+1)}{2}$$

$$= N + N^2$$

$$= O(N^2)$$

```
for( i=1; i<=N; i++)
{
    for(j=1; j<=N; j*=2)
    {
        =
    }
}
```

$N \cdot (1 + \log_2(N))$

$\downarrow$

$O(N \cdot \log(N))$

i	j	# itr
1	1 to N (*2)	$1 + \log_2(N)$
2	"	"
3	"	"
4	"	"
⋮	⋮	⋮
N	"	"

$N(1 + \log_2(N))$

```
i = 1
while ( i <= N)
{
    j = 1
    while ( j <= N)
    {
        j++
    }
    i++
}
```

i	j	# ito
1	1 to N	1 + N
2	1 to N	1 + N
3	⋮	1
4	⋮	1
⋮	⋮	1
N	1 to N	+ 1 + N
		N + N²

$O(N^2)$

```
i = 1 , j = 1
while ( i <= N)
{
    while ( j <= N)
    {
        j++
    }
    i++
}
```

i	j	#ib
1	1 to N	1 + N
2	———	1
3	——→	1
4	———	1
⋮		
N	———	+ 1
		2N

TC : O(N)

COMPARING IDEAS / APPROACHES

Sorting



Tanny

20 sec (Python)

↓

8 secs (Rust) (Macbook pro)

↓

6 secs Volcano



Mr Beast

12 secs (Rust)

=

↓ (Window xp)

6 secs Macbook pro

Mark E

BIG O TIME COMPLEXITY

⇐ # of iterations

1) Count of iterations → $2N + 100 + N^2 + \log_2(N)$

2) Remove all the constants → $N + N^2 + \log_2(N)$

3) Keep only the most dominating term / remove all small order terms → N^2

N

$\log_2(N)$

1

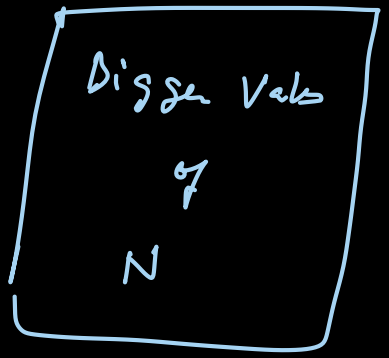
65

728

$2N + 100 + N^2 + \log_2(N) = O(N^2)$

$N^3 + N + N \log(N) + N^2$ $O(N^3)$

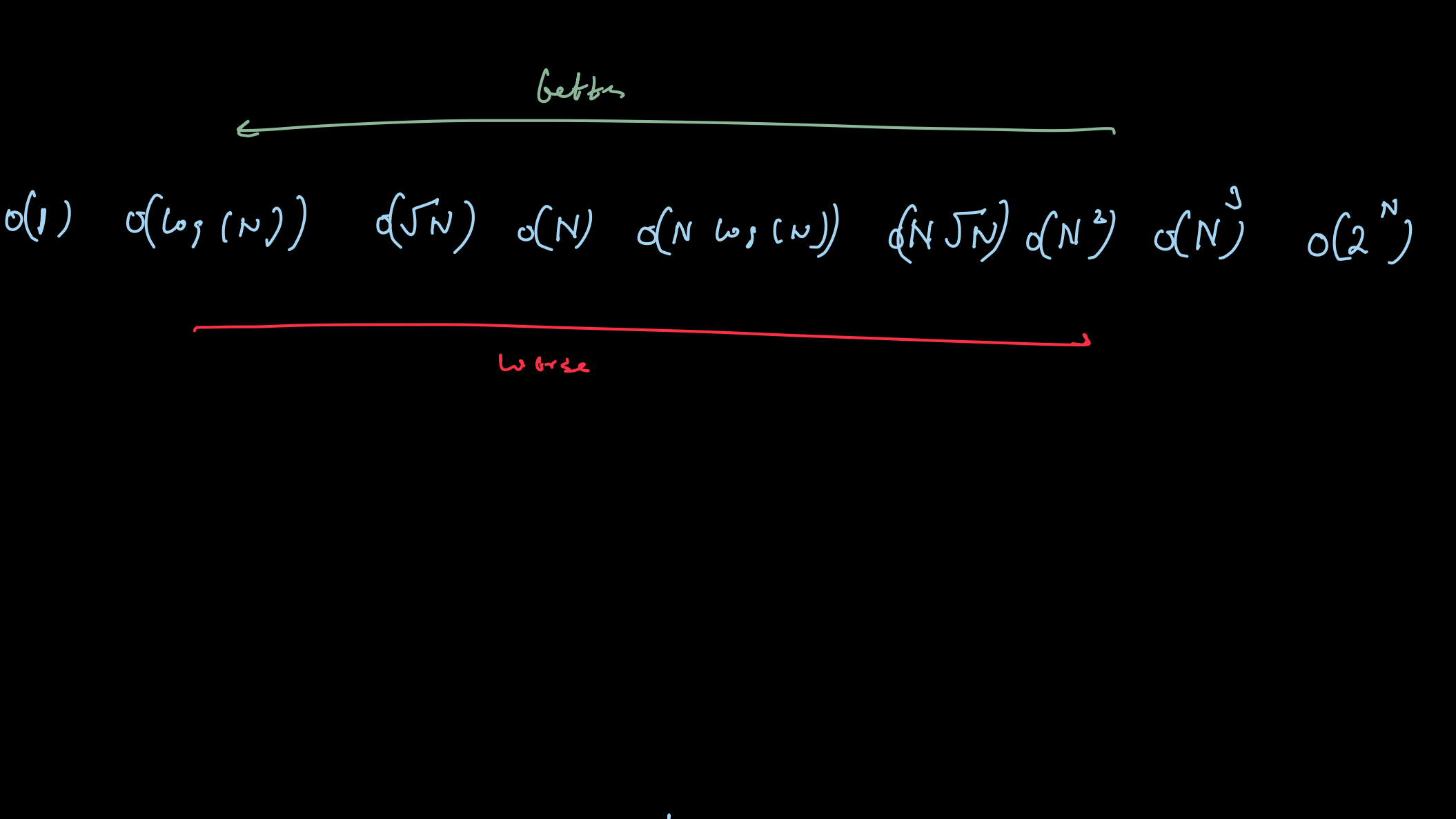
$N^2 \log N + N^2 \sqrt{N} + N^3$ $O(N^3)$



$N = 10^{18}$

$\sqrt{N} = 10^9$

$\log_2(10^{18}) \sim 60$


better
O(1) O(log(N)) O(√N) O(N) O(N log(N)) O(N √N) O(N²) O(N³) O(2^N)
worse

(1)

```
int sum(int N)
{
    return N*(N+1)/2;
}
```

$O(1)$

< 1 secs

$N \leq 10^9$

(2)

```
int sum(int N)
{
    ans = 0
    for(i=1; i<=N; i++)
    {
        ans += i
    }
    return ans
}
```

$O(N)$

10 secs

$N = 10^9$

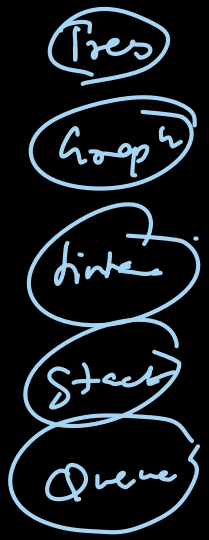
BIG O SPACE COMPLEXITY

MLE

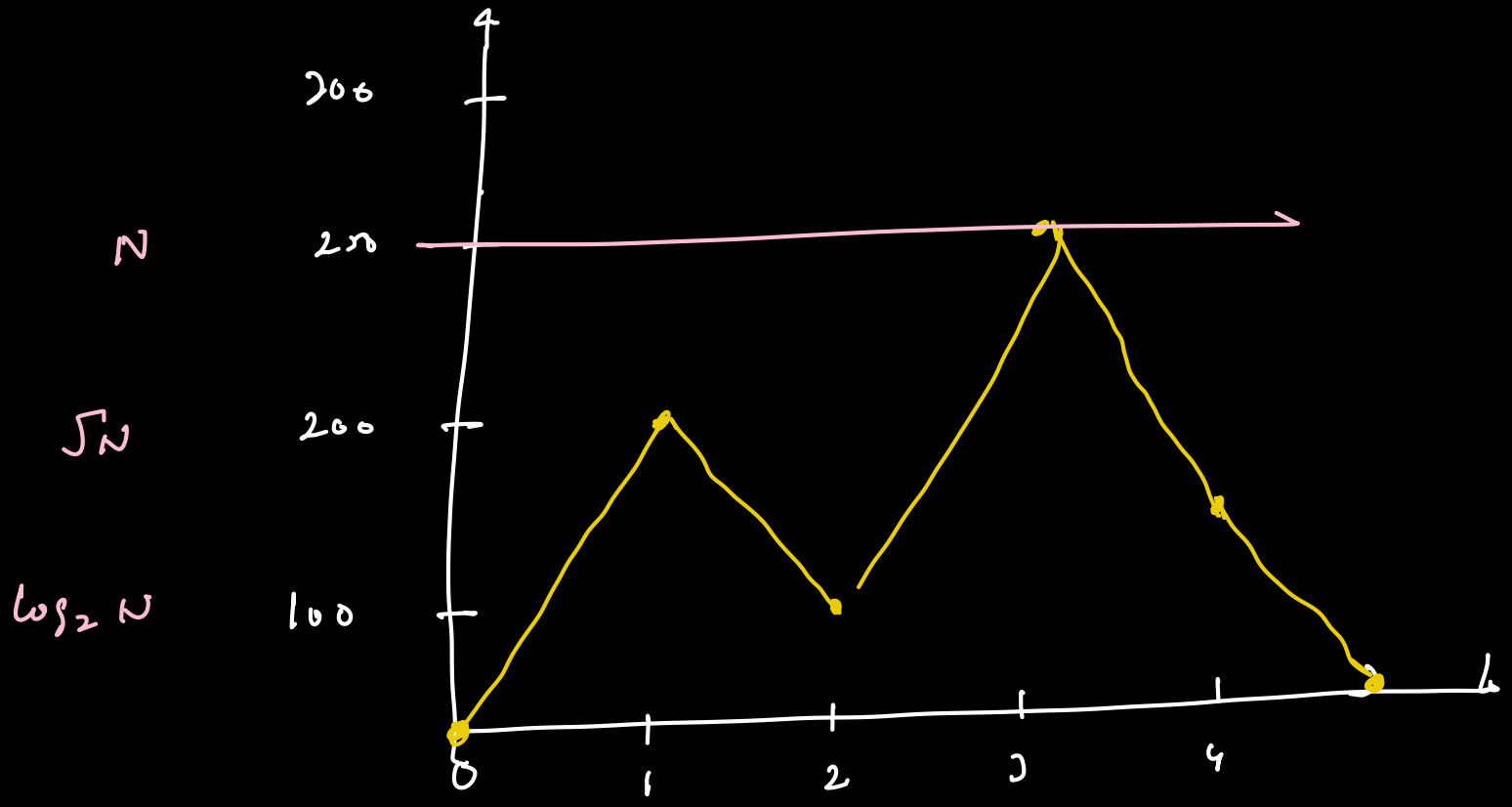
↓

Memory limit Exceeded

int A[N] → $10^6 - 10^7$



Max Space Utilised at any point of execution.



$0 + 200 + 100 + 250 + 150 + 0$?!

$\max(0, 200, 100$